

ICHEON, Korea, Republic of

WMO: 470970

Lat: 37.2639N

Lon: 127.4842E

Elev: 79

StdP: 100.38

Time Zone: 9.00 (E09)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.3	-12.0	-23.6	0.5	-8.4	-21.6	0.5	-8.1	4.9	-1.8	4.4	-2.1	1.1	270	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.2	32.9	24.3	31.6	23.6	30.2	22.5	25.8	29.9	25.2	29.4	24.5	28.6	1.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.8	20.0	27.7	24.0	19.1	27.6	23.2	18.1	27.0	80.4	29.9	77.5	29.3	74.7	28.8	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.0	4.3	3.7	DB	-17.3	34.9	3.3	1.4	-19.7	35.9	-21.6	36.7	-23.5	37.5	-25.9	38.5	(4)
(5)			WB	-17.8	26.8	3.1	1.1	-20.0	27.5	-21.9	28.2	-23.6	28.8	-25.8	29.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	-3.8	-0.2	5.6	11.6	18.0	22.6	25.3	25.7	20.5	13.8	6.2	-1.7	(6)	
	DBStd	10.75	4.30	3.87	3.66	3.44	2.56	1.90	2.05	2.59	2.60	3.44	4.43	4.23	(7)	
	HDD10.0	1382	427	286	146	23	0	0	0	0	0	9	129	362	(8)	
	HDD18.3	2978	685	519	396	201	37	1	0	0	9	146	364	620	(9)	
	CDD10.0	2127	0	0	9	72	248	377	476	488	316	125	16	0	(10)	
	CDD18.3	682	0	0	0	1	28	128	217	229	75	4	0	0	(11)	
	CDH23.3	5578	0	0	0	25	325	946	1763	2069	427	22	0	0	(12)	
	CDH26.7	1720	0	0	0	2	58	246	565	782	67	1	0	0	(13)	
(14) Wind	WSAvg	1.3	1.2	1.4	1.8	1.8	1.5	1.4	1.3	1.2	1.0	1.0	1.2	1.2	(14)	
(15) Precipitation	PrecAvg	1311	25	28	50	97	98	143	339	251	159	43	48	28	(15)	
	PrecMax	2313	74	128	123	260	266	456	696	594	710	210	144	56	(16)	
	PrecMin	949	0	0	10	7	23	12	99	72	6	1	6	3	(17)	
	PrecStd	311	22	31	28	69	60	112	162	139	150	43	36	16	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	12.9	20.6	26.3	30.1	31.7	34.1	35.1	30.1	25.6	20.5	10.4	(19)	
		MCWB	4.1	7.6	11.8	13.8	16.4	19.2	24.5	25.5	22.6	18.5	14.8	7.1	(20)	
	2%	DB	6.0	10.7	17.9	23.4	28.2	30.2	32.4	33.7	28.4	23.4	17.0	8.1	(21)	
		MCWB	2.1	5.4	9.7	12.4	15.9	19.3	24.0	25.1	20.5	16.3	12.0	4.8	(22)	
	5%	DB	4.2	8.4	15.2	21.5	26.7	29.3	31.1	32.2	27.2	21.6	15.0	6.5	(23)	
		MCWB	0.9	3.6	8.1	11.4	15.6	19.1	23.7	24.6	19.6	15.1	10.8	3.6	(24)	
	10%	DB	2.5	6.1	12.9	19.3	25.1	28.1	29.8	30.8	25.9	20.1	13.2	4.7	(25)	
		MCWB	-0.5	2.0	6.8	10.5	14.9	18.9	23.2	24.0	19.1	14.6	9.9	2.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	10.2	12.9	15.2	20.3	22.7	26.4	27.1	24.8	20.5	15.5	8.5	(27)	
		MCDB	7.9	12.0	17.2	22.8	22.0	26.4	29.9	31.8	28.1	23.1	18.8	9.5	(28)	
	2%	WB	2.8	5.9	11.1	14.1	18.1	21.7	25.5	26.2	23.0	18.3	13.2	5.6	(29)	
		MCDB	5.3	9.8	15.8	20.5	24.3	26.2	29.1	30.7	25.3	22.1	16.3	7.3	(30)	
	5%	WB	1.2	4.0	9.3	12.8	17.1	21.1	24.9	25.6	21.7	16.4	11.6	3.9	(31)	
		MCDB	3.5	7.7	14.0	18.7	23.4	25.8	29.1	30.0	25.1	19.7	14.3	6.0	(32)	
	10%	WB	-0.1	2.6	7.4	11.6	16.3	20.3	24.3	24.9	20.6	15.0	10.1	2.0	(33)	
		MCDB	2.3	5.8	12.1	17.2	22.4	25.4	28.2	29.2	24.1	18.8	12.6	4.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.1	11.6	12.6	12.5	10.4	7.3	8.2	10.1	12.1	9.9	8.8	(35)	
		MCDBR	10.2	13.1	15.4	16.8	15.7	12.5	9.7	10.2	11.8	13.3	11.7	9.9	(36)	
	5% WB	MCWBR	7.0	8.3	8.2	7.4	5.5	3.9	3.1	3.4	4.8	6.4	7.0	7.2	(37)	
		MCWBR	9.2	11.7	13.1	13.1	11.5	8.7	7.2	8.7	8.1	9.9	10.0	8.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.425	0.476	0.561	0.599	0.610	0.677	0.643	0.568	0.492	0.480	0.462	0.420	(40)	
		taud	1.971	1.881	1.731	1.693	1.714	1.638	1.764	1.948	2.088	1.996	1.982	2.005	(41)	
	Ebn at Noon		712	726	703	707	710	662	682	725	759	716	668	684	(42)	
		Edh at Noon	136	169	215	236	235	253	222	181	149	148	134	124	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.48	3.24	4.18	4.88	5.50	5.11	3.99	4.21	4.01	3.54	2.45	2.21	(44)	
	RadStd		0.15	0.26	0.44	0.38	0.48	0.46	0.58	0.60	0.46	0.26	0.25	0.16	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+89	+49

Nomenclature: See separate page